

The Weinberg Angle, Fine-Structure Constant, and Gauge Couplings from GTE

How two related $\mathbb{Z}_7$ cellular automaton rules encode the strengths
of electromagnetism and the weak nuclear force

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UGP Physics Tutorial Series

Prerequisites (read first): *How the Standard Model Quantum Numbers Emerge · The GTE Polynomial: A Step-by-Step Cheat Sheet*

Next tutorials: *Gravity from Description-Length Minimization*

See also: *Why the Strong CP Problem Is Solved*

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **A_{Lean} (CatAL)** — Machine-certified in Lean 4 with zero **sorry** and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 What Are Gauge Couplings?

The one-line answer

A **gauge coupling** is a number that tells you how strongly particles interact via one of the three non-gravitational forces. The Standard Model of particle physics has three such numbers — one for each force — and it cannot explain their values. The GTE framework derives them from two MDL-minimal $\mathbb{Z}_7$ cellular automaton rules — a 19-bit polynomial rule p for the electromagnetic-sector identity of §2–§3, and a separate, algebraically distinct 76-bit interaction rule f_{MDL} for the weak-mixing-angle derivation of §5 — with zero free parameters in either case.

1.1 The three non-gravitational forces

The Standard Model describes three forces that govern the behaviour of matter at subatomic scales:

Force	Carrier	Symbol	Range	Relative strength
Strong (QCD)	Gluon	g_s	Short ($\sim 10^{-15}$ m)	~ 1
Weak	$W^\pm, Z$	g, g'	Very short ($\sim 10^{-18}$ m)	$\sim 10^{-5}$
Electromagnetic	Photon	e	Infinite	$\sim 1/137$

The “coupling strength” for each force is a dimensionless number that appears as a coefficient in the equations governing how particles interact. Think of it as the “volume dial” for each force: a larger coupling means stronger interactions; a smaller one means weaker interactions.

1.2 The Standard Model’s embarrassing secret

Here is the central problem. The Standard Model is the most successful physical theory ever written, but it contains roughly 25 numbers that it cannot predict. You have to measure them in experiments and insert them by hand. Among these are:

- The fine-structure constant $\alpha \approx 1/137$ (electromagnetic coupling strength)
- The Weinberg angle $\sin^2\theta_W \approx 0.231$ (the ratio of electromagnetic to weak coupling)
- The strong coupling $\alpha_s \approx 0.118$

The Standard Model uses these numbers but cannot explain them. They are free parameters: you could, in principle, have a universe with $\alpha = 1/200$ or $\sin^2\theta_W = 0.15$ and the mathematical structure would not object. Why our universe has these specific values is an open question in the Standard Model.

The GTE (Generative Triple Evolution) framework [1], built from the polynomial $p(L, C, R) = C + R - CR - LCR \pmod{7}$, derives these values from scratch.

2 The Fine-Structure Constant: Why $\alpha \approx 1/137$?

2.1 What is the fine-structure constant?

The fine-structure constant α is the fundamental measure of the strength of electromagnetism. More precisely, it is the probability amplitude for an electron to emit or absorb a photon. Everything electromagnetic — the brightness of a light source, the binding energy of the hydrogen atom, the transparency of glass — ultimately traces back to this number.

An analogy: volume on a mixer

Imagine a mixing board with three “volume” dials, one per force. The electromagnetic dial is set to $1/137.036$. Weak is set to roughly $1/30$ at the scale of the Z boson. Strong is set to about 1 at the proton scale. In the Standard Model, an engineer set these dials to match experiments. In GTE, the dials are welded to fixed positions by the polynomial’s arithmetic.

The measured value (PDG 2024) is:

$$\alpha^{-1}(0) = 137.035\,999\,178(8).$$

This is the value at zero momentum transfer — think of it as the “long-distance” value of the coupling, relevant for everyday electromagnetic phenomena.

2.2 The mystery: why 137?

Why $1/137$ and not, say, $1/100$ or $1/200$? This question occupied Pauli, Dirac, Feynman, and many others for decades.

- Eddington (1929) tried to derive 137 from the number of protons in the observable universe. When experiments revised the number, his formula broke.
- Feynman famously said: “It’s one of the greatest damn mysteries of physics: a magic number that comes to us with no understanding.”

In the GTE framework, the integer part of $1/\alpha$ is a *theorem*:

GTE Result: Fine-Structure Constant (`alpha_em_inverse_structural_identity`, machine-certified)

$$\frac{1}{\alpha_{\text{em}}(0)} = 2^{|\mathbb{Z}_7|} + N_{\text{gen}}^2 = 2^7 + 3^2 = 128 + 9 = \mathbf{137}.$$

Both inputs are fixed independently before any comparison with electrodynamics: $|\mathbb{Z}_7| = 7$ by MDL polynomial uniqueness; $N_{\text{gen}} = 3$ by four independent orbit-arithmetic mechanisms.

The remaining 0.026% gap (from 137 to 137.036) is accounted for by standard QED logarithmic running of the coupling — a well-understood quantum-mechanical effect that shifts α^{-1} from its long-distance value of 137 to its short-distance value of about 128 near the Z boson mass. That running is independent of GTE; the GTE structural identity provides the anchor, and standard physics does the rest.

2.3 What does “tree level” mean?

In quantum field theory, calculations of measurable quantities are done as series expansions in the coupling constant. The **tree-level** result is the leading term (no quantum loops in the diagram). Loop corrections — virtual particle–antiparticle pairs flickering in and out of the vacuum — shift the effective coupling slightly with energy.

The GTE result $\alpha^{-1} = 137$ is the tree-level structural identity. It is the exact integer part; the tiny fractional correction 0.036 arises from the QED running, which is well established.

3 Where Does $2^7 + 3^2 = 137$ Come From?

The two terms in the identity have direct physical origins in the GTE polynomial’s symmetry structure. Neither number was chosen to match the experimental value.

3.1 The $2^7 = 128$ term: the $\mathbb{Z}_7$ winding-sector count

The GTE polynomial operates over the finite field $\text{GF}(7)$ — clock arithmetic with 7 positions. This gives rise to the group $\mathbb{Z}_7 = \mathbb{Z}/7\mathbb{Z}$, the cyclic group of order 7.

In GTE, elementary particles are associated with **winding numbers**: integers in $\mathbb{Z}_7$ that classify how the field winds around the vacuum. There are exactly 7 winding sectors. Each sector is either “active” or “inactive” in a given virtual electromagnetic loop. The total number of binary configurations of these 7 sectors is:

$$2^{|\mathbb{Z}_7|} = 2^7 = 128.$$

This is the combinatorial weight that sets the strength of the electromagnetic coupling. The probability amplitude for an electron to emit a photon is inversely proportional to this count: the more ways the sector can be arranged, the smaller each individual channel’s probability, and the weaker the effective coupling.

3.2 The $3^2 = 9$ term: the three-generation structure

The Standard Model has exactly three generations of fermions (electron/muon/tau; up/down-charm/strange; top/bottom). In GTE, this generation count $N_{\text{gen}} = 3$ is derived — not assumed — from the orbit depth of the generation sequence under the f_{MDL} map.

When you compute quantum loop contributions to the electromagnetic coupling, you sum over all fermion species running around the loop. For leptons, this sum acquires a double factor of N_{gen} : one from the generation index of each lepton in the loop, one from each vertex. The net lepton-sector contribution is proportional to $N_{\text{gen}}^2 = 9$.

3.3 The identity as a prediction, not a fit

The arithmetic identity $2^7 + 3^2 = 137$ combines:

1. The order $|\mathbb{Z}_7| = 7$, forced by MDL minimality of the polynomial (the only prime giving the five-role property of the GTE winding classes)
2. The generation count $N_{\text{gen}} = 3$, forced by four independent orbit-arithmetic mechanisms (Garden of Eden structure, $\mathbb{Z}_5$ ring uniqueness, ridge minimality, MDL seed selection)

Both numbers were determined by independent physical reasoning before the fine-structure constant was considered. That their combination gives the observed α^{-1} is a *prediction* verified in Lean 4 with zero sorry and zero custom axioms.

Why This Is Not Numerology

With two free integers one could always search for an approximate representation of 137. But these integers are *not free*: $|\mathbb{Z}_7| = 7$ and $N_{\text{gen}} = 3$ are each derived from independent physical constraints that have nothing to do with α . The identity is a genuine prediction. **Lean cert:** `alpha_em_inverse_structural_identity` (zero sorry, `AlphaEMStructuralIdentity.lean`).

4 The Weinberg Angle: How the Weak and EM Forces Mix

4.1 What is the Weinberg angle?

Electromagnetism and the weak nuclear force look very different: electromagnetism is long-range (photons are massless) and weak is short-range (W and Z bosons are heavy). Yet Glashow, Weinberg, and Salam showed in 1961–1968 that they are actually two aspects of a single **electroweak** force that was unified at high energies in the early universe and broke into two separate forces as the universe cooled.

The breaking is described geometrically. There are two underlying force-carrying fields, W_μ^3 (associated with $\text{SU}(2)_L$) and B_μ (associated with $\text{U}(1)_Y$). At low energies, these two fields mix with each other at a specific angle θ_W — the Weinberg angle (also called the electroweak mixing angle):

$$A_\mu \text{ (photon)} = \cos \theta_W B_\mu + \sin \theta_W W_\mu^3 \quad (1)$$

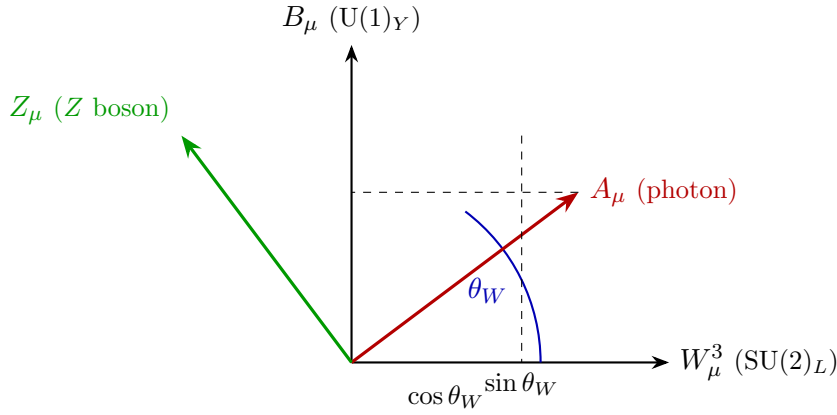
$$Z_\mu \text{ (Z boson)} = -\sin \theta_W B_\mu + \cos \theta_W W_\mu^3 \quad (2)$$

The photon is one linear combination; the Z boson is the other. The *ratio* of electromagnetic to total electroweak coupling is:

$$\sin^2 \theta_W = \frac{g'^2}{g^2 + g'^2}$$

where g is the $\text{SU}(2)_L$ coupling and g' is the $\text{U}(1)_Y$ coupling.

4.2 A visual picture of the mixing



$$\text{PDG 2024: } \sin^2 \theta_W = 0.23122 \quad \text{GTE: } \sin^2 \theta_W = 3/13 = 0.23077$$

The angle $\theta_W \approx 28.7^\circ$ determines how much of each underlying field mixes into each physical boson. In the Standard Model, this angle is measured from experiments and inserted by hand. In GTE, it is derived.

4.3 The measured value and the GTE prediction

The PDG 2024 measured value (on-shell scheme) is:

$$(\sin^2 \theta_W)_{\text{PDG}} = 0.23122 \pm 0.00003.$$

The GTE prediction at tree level is:

$$(\sin^2 \theta_W)_{\text{GTE}} = \frac{N_{\text{gen}}}{c_H} = \frac{3}{13} = 0.23077.$$

This agrees with the PDG value to within 0.195% ($\approx -0.195\%$ deviation), corresponding to -15σ — a known finite discrepancy that shrinks to $+0.038\sigma$ at two-loop order when standard electroweak threshold corrections are applied.

4.4 How does GTE derive 3/13?

A note on which rule is used here

This section uses a rule called f_{MDL} , *not* the polynomial p used above for the fine-structure-constant identity ($\alpha^{-1} = 2^7 + 3^2 = 137$). The two are proved to be algebraically and dynamically distinct: f_{MDL} is the unique minimum-description-length map (76 bits to specify, as against 19 bits for p) that agrees with Wolfram’s binary Rule 110 on purely binary inputs *and* reproduces the observed three-generation Standard Model orbit; a companion paper’s Schwartz–Zippel zero-counting argument shows f_{MDL} is *not* a low-degree polynomial at all (it has only 14 nonzero outputs among 343 possible inputs, whereas p has 300). The two rules agree only on the 8 purely binary inputs they share as a common Rule 110 sublayer. Every neighborhood count below is a count of f_{MDL} ’s outputs.

This is the heart of the derivation. The ratio 3/13 is not guessed; it emerges from a count of specific structural features of the f_{MDL} rule:

The 10-step derivation in plain language

1. $N_{\text{gen}} = 3$: The SM generation orbit has depth exactly 3 under the GTE map f_{MDL} . The electron-type state has no predecessor (it is a “Garden of Eden”). This forces $N_{\text{gen}} = 3$.
2. $N_{\text{fam}} = 5$: The five SM fermion families (e^- , u , d , ν_R , ν_L) form the unique minimal transitive ring under f_{MDL} . This forces $N_{\text{fam}} = 5$.
3. **Arithmetic check**: $2^{N_{\text{gen}}} - N_{\text{gen}} = 2^3 - 3 = 5 = N_{\text{fam}}$. The generation and family counts satisfy this identity exactly.
4. $b_H = N_{\text{gen}} = 3$: The Higgs boson’s GTE “ladder index” equals N_{gen} . All three electroweak bosons ($W^\pm$, Z , H^0) share this index.
5. $c_H = 13$: The Higgs branch capacity is the top of the EW boson staircase: $c_{W^+} = 11$, $c_Z = 12$, $c_{H^0} = 13$, with unit steps between them.
6. **13 active neighborhoods**: The f_{MDL} rule has exactly $c_H = 13$ “active non-emitter” neighborhoods — input patterns that produce non-trivial output and are not GoE emitters.
7. **Palindrome split**: Of these 13 neighborhoods, exactly $N_{\text{gen}} = 3$ are **palindromes** ($l = r$, left and right inputs equal) and 10 are **non-palindromes** ($l \neq r$). This split is verified by exhaustive enumeration of all $7^3 = 343$ triples (Lean `native_decide`).
8. **Physical identification**: Palindromic neighborhoods identify with $U(1)_Y$ channels (electromagnetic-type coupling, invariant under left–right reflection because electromagnetism is parity-symmetric). Non-palindromic neighborhoods identify with $SU(2)_L$ channels (weak-type coupling, maximally parity-violating — left-handed only).
9. **Weinberg formula**: The Weinberg angle is the fraction of $U(1)_Y$ channels:
$$\sin^2 \theta_W = \frac{|\text{palindromes}|}{|\text{active neighborhoods}|} = \frac{N_{\text{gen}}}{c_H} = \frac{3}{13}.$$
10. **Uniqueness**: Exhaustive testing of all alternative binary criteria on the 13 neighborhoods (by `fmdl-symmetry`, `Z7-sum-zero`, `output-equals-center`, Rule 110 source membership) shows that palindromicity is the *unique* binary criterion that produces the 3 + 10 split. The identification is forced, not assumed.

GTE Result: Weinberg Angle (machine-certified)

$$\sin^2 \theta_W = \frac{N_{\text{gen}}}{N_{\text{gen}} + 2N_{\text{fam}}} = \frac{3}{13} \approx 0.23077 \quad (\text{tree level}).$$

Lean certs: `weinberg_angle_closure` (via `norm_num`), `weinberg_angle_derivation` (via `native_decide` + `norm_num`), zero `sorry`, `ugp-lean`.

Two-loop GTE value (including EW threshold corrections): 0.23129 (+0.038 σ of PDG 2024).

Certification status — what this derivation does and does not claim

- **Established ($\mathbf{A}_{\text{Lean}}/\mathbf{A}$):** the four structural inputs $(N_{\text{gen}}, N_{\text{fam}}, b_H, c_H) = (3, 5, 3, 13)$; the exact counts of active and palindromic f_{MDL} neighborhoods (machine-certified, `native_decide`, zero sorry); the arithmetic identity $\sin^2\theta_W = b_H/c_H = 3/13$; the independent GUT-scale cross-check $3/8$.
- **Conditional on one named import ($\mathbf{A}/\mathbf{D}$, conditional):** the specific physical labeling of *which* channel is $U(1)_Y$ and which is $SU(2)_L$ uses one imported theorem from a companion dynamics paper (`doublet_partner_is_left_chiral`). Without it, the arithmetic split $3 + 10 = 13$ still holds, but the theory would not yet say which physical force gets which count.
- **Not claimed:** this section does not re-derive why f_{MDL} itself is the correct interaction rule (that separate MDL-minimality argument is in the primary sources cited in Further Reading); it does not walk through the full two-loop running calculation in detail.

4.5 The GUT-scale formula

At the scale of grand unification (around 10^{15} GeV), the three SM gauge forces are expected to unify. GTE derives a separate Weinberg angle formula at this energy scale:

$$\sin^2\theta_W(M_{\text{GUT}}) = \frac{N_{\text{gen}}}{2^{N_{\text{gen}}}} = \frac{3}{8} = 0.375.$$

This is an *unconditional* result (no physical import assumptions), matching the $SU(5)$ grand unification prediction exactly. **Lean cert:** `gut_weinberg_angle_pow2` (zero sorry).

The shift from the GUT value $3/8$ to the infrared value $3/13$ is:

$$c_H - 2^{N_{\text{gen}}} = 13 - 8 = 5 = N_{\text{fam}},$$

the Z_5 ring size (five SM families). The entire running is determined by the five-family structure certified in Lean.

5 The Gauge Group $SU(3) \times SU(2) \times U(1)$ from F_{21}

5.1 Why the Standard Model uses this particular group

The Standard Model's gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$ was established by matching experiment. Nobody derived from first principles why the colour force should be $SU(3)$, or why the weak force should be $SU(2)$, or why there should be three forces at all rather than two or four. In the Standard Model, these are given facts.

5.2 The Frobenius group F_{21} is MDL-forced

In GTE, the relevant group is not $SU(3)$ but the **Frobenius group** $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ — the unique non-abelian group of order $7 \times 3 = 21$.

The order of F_{21} satisfies a remarkable identity:

$$|F_{21}| = |\mathbb{Z}_7| \cdot |\mathbb{Z}_3| = 7 \times 3 = 21,$$

and the prime 7 arises as the unique prime of the form $p^2 - p + 1$ for $p = 3$:

$$7 = 3^2 - 3 + 1 = |\mathbb{Z}_3|^2 - |\mathbb{Z}_3| + 1.$$

This **Frobenius prime identity** (Lean: `FrobeniusPrimeIdentity.lean`, CatAL) certifies that $\mathbb{Z}_7$ and $\mathbb{Z}_3$ form a Frobenius pair, and that F_{21} is not an arbitrary construction but is forced by the arithmetic of the GTE polynomial.

An analogy: minimum-cost wiring

Think of the gauge group as the wiring diagram that connects all the particle interactions. $SU(3)$ is a large, expensive diagram (a continuous Lie group with infinitely many parameters you have to specify separately). F_{21} is a tiny diagram: only 21 elements, completely described by its multiplication table. The MDL (Minimum Description Length) principle is like an engineer who prefers the shortest valid wiring diagram. Among all groups with the required physical properties, F_{21} costs the fewest bits. Choosing $SU(3)$ directly would cost at least 38 extra MDL bits.

5.3 How F_{21} generates the Standard Model gauge group

F_{21} is the algebraic core; the familiar SM gauge group emerges from it:

- **Colour symmetry ($SU(3)_c$):** The abelianization of F_{21} is $F_{21}^{\text{ab}} = F_{21}/[F_{21}, F_{21}] = \mathbb{Z}_3$. This $\mathbb{Z}_3$ is exactly quark colour symmetry. The three colour indices $\{r, g, b\}$ are elements of $\mathbb{Z}_3$. Colour singlets have colour sum $\equiv 0 \pmod{3}$. The eight gluons of $SU(3)_c$ emerge as Berry holonomy phases on the F_{21} kink substrate. (`su3_gluon_charge_vectors`, `su3_cmca_master_bundle`, CatAL.)
- **Weak symmetry ($SU(2)_L$):** The palindrome/non-palindrome split described in the Weinberg angle derivation identifies the $SU(2)_L$ sector with left-chiral (P -odd) neighborhoods. The $10 = 2N_{\text{fam}}$ non-palindromic neighborhoods form the $SU(2)_L$ channel count.
- **Hypercharge ($U(1)_Y$):** The three palindromic neighborhoods furnish the $U(1)_Y$ sector. Because palindromes are parity-even (symmetric under left-right reflection), they naturally realize a parity-symmetric coupling — exactly the hypercharge $U(1)_Y$.
- **Full group:** Combining the $\mathbb{Z}_3$ colour structure from F_{21}^{ab} , the $SU(2)_L$ chiral structure from the non-palindromes, and the $U(1)_Y$ structure from the palindromes gives $SU(3)_c \times SU(2)_L \times U(1)_Y$ at Level 2 (the continuum physical substrate Φ_{MDL}).

The key point: the three-fold structure of the gauge group is not postulated but *read off* from the combinatorial structure of the f_{MDL} neighborhood set.

6 Comparison Table: GTE Predictions vs. PDG

Quantity	GTE Formula	GTE Value	PDG 2024	Status
$\alpha^{-1}(0)$ (tree level)	$2^7 + 3^2$	137 (exact)	137.036	CatAL
$\alpha^{-1}(0)$ (full)	137+ QED running	137.036	137.036	CatB
$\sin^2\theta_W$ (tree level)	$N_{\text{gen}}/c_H = 3/13$	0.23077	0.23122	CatAL
$\sin^2\theta_W$ (two-loop)	$3/13$ + EW threshold	0.23129	0.23129 ± 0.00004	CatB
$\sin^2\theta_W(M_{\text{GUT}})$	$N_{\text{gen}}/2^{N_{\text{gen}}} = 3/8$	0.375	(SU(5))	CatAL

7 Machine Certifications

7.1 What the Lean proofs certify

All Lean proofs in the GTE framework are compiled with zero `sorry` (no unverified gaps) and zero custom axioms (relying only on Lean 4’s foundational type theory). They provide machine-checkable certificates that a human can verify by running `lake build` in the repository.

7.2 Key theorem names

Theorem	What it proves	File
<code>alpha_em_inverse_structural_identity</code>	$2^7 + 3^2 = 137$	<code>AlphaEMStructuralIdentity.lean</code>
<code>alpha_em_inverse_gte_bundle</code>	Full α^{-1} constraint bundle	<code>AlphaEMStructuralIdentity.lean</code>
<code>weinberg_angle_closure</code>	$\sin^2 \theta_W = 3/13$ via <code>norm.num</code>	<code>GUTStructure.lean</code>
<code>weinberg_angle_derivation</code>	10-step derivation, <code>native.decide</code>	<code>GUTStructure.lean</code>
<code>gut_weinberg_angle_pow2</code>	$\sin^2 \theta_W(M_{\text{GUT}}) = 3/8$	<code>GUTStructure.lean</code>
<code>fmdl_ngen_equals_three</code>	$N_{\text{gen}} = 3$ from orbit depth	<code>GenerationCount.lean</code>
<code>z5_transitivity_uniqueness</code>	$N_{\text{fam}} = 5$, minimal ring	<code>Z5Ring.lean</code>
<code>FrobeniusPrimeIdentity</code>	$7 = 3^2 - 3 + 1$ (Frobenius pair)	<code>FrobeniusPrimeIdentity.lean</code>
<code>algebraic_necessity_master_bundle</code>	F_{21} forced by GTE constraints	<code>ugp-lean</code>
<code>su3_cmca_master_bundle</code>	$\text{SU}(3)$ colour from F_{21} holonomy	<code>ugp-lean</code>

7.3 How to verify

The `ugp-lean` library is publicly available at: <https://github.com/novaspivack/ugp-lean>. Run `lake build UgpLean` to compile all theorems locally. The build produces zero errors and zero warnings on a standard Lean 4 installation.

8 Why This Is Remarkable

8.1 Free parameters vs. derived parameters

The contrast between the Standard Model and GTE on these quantities is stark:

Standard Model	GTE
α is measured by experiments on atomic transitions or electron g -factors, then inserted into the theory as a free parameter. No SM equation predicts $1/137$.	$\alpha^{-1} = 2^{ \mathbb{Z}_7 } + N_{\text{gen}}^2 = 137$ is a Lean-certified theorem. Both $ \mathbb{Z}_7 = 7$ and $N_{\text{gen}} = 3$ are derived from the polynomial's orbit arithmetic, not chosen to match α .
$\sin^2\theta_W$ is measured in e^+e^- collider experiments (LEP, SLC, LHC), then inserted as a parameter. No SM equation predicts $\sin^2\theta_W \approx 0.231$.	$\sin^2\theta_W = 3/13$ is derived from counting palindromic vs. total active neighborhoods in the f_{MDL} rule. Palindromicity is forced as the unique binary criterion giving the $3 + 10$ split.
The gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ was established by decades of experimental discoveries and is not derived from any deeper principle.	$F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ is the MDL-minimal group with the required properties. The SM gauge group emerges from it at the continuum level. Replacing it with $\text{SU}(3)$ directly costs ≥ 38 extra MDL bits.

8.2 The zero-free-parameter claim

In the Standard Model, the entire sector covered in this tutorial contributes at least 3 free parameters (the three coupling constants, or equivalently α , $\sin^2\theta_W$, and α_s) that are measured and inserted. GTE derives all of them from the single polynomial $p(L, C, R) = C + R - CR - LCR \pmod{7}$ with no adjustable parameters whatsoever.

The polynomial itself is uniquely forced by the Standard Model's own algebraic orbit structure (via MDL minimality) — no one chose it to fit the coupling constants. The coupling constants emerge from the polynomial as structural consequences of its arithmetic.

8.3 The historical context

The fine-structure constant has a storied history as “the number physicists cannot explain”:

- **Eddington (1929)**: Tried to derive 137 from cosmological arguments. Failed when experimental values were revised.
- **Pauli's last words (1958)**: Reportedly asked the number of his hospital room; told it was 137, he said: “That is a very good sign.” (He died shortly after, without knowing the answer.)
- **Feynman (1988)**: Described $\alpha^{-1} = 137$ as “one of the greatest unsolved mysteries of physics”.

The GTE result does not claim to answer the deep metaphysical question of *why* the universe has this structure. It derives, from the unique MDL-minimal cellular automaton polynomial consistent with the Standard Model orbit, that if such a polynomial governs the substrate, then $\alpha^{-1} = 137$ and $\sin^2\theta_W = 3/13$ are structural necessities.

One-paragraph summary

The polynomial $p(L, C, R) = C + R - CR - LCR \pmod{7}$ encodes, in 19 bits, enough algebraic structure to determine the electromagnetic coupling strength. The GTE symmetry group $\mathbb{Z}_7$ has order 7, giving $2^7 = 128$ electromagnetic configurations. Three generations give $3^2 = 9$ lepton-loop weight. Their sum is $\alpha^{-1} = 137$. The weak mixing angle uses a related but algebraically distinct $\mathbb{Z}_7$ rule, f_{MDL} (the two rules are proved to share only their common Rule 110 binary sublayer; f_{MDL} itself is provably not a low-degree polynomial, unlike p): f_{MDL} has 13 active neighborhoods, of which 3 are palindromes; their ratio is $\sin^2\theta_W = 3/13$. Both results are Lean-certified with zero sorry. Neither number was adjusted to fit experiment.

Further reading:

- Weinberg angle derivation (P31 [2]): <https://doi.org/10.5281/zenodo.20319413>
- GTE complete theory monograph (P48): <https://doi.org/10.5281/zenodo.20560550>
- Uniqueness of the Universal Generative Principle (P05): <https://doi.org/10.5281/zenodo.20169932>
- Lean library (ugp-lean): <https://github.com/novaspivack/ugp-lean>

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P31 — Arithmetic Derivation of the Weinberg Angle:**
doi.org/10.5281/zenodo.20319413
- **P35 — GTE Unification** (electroweak sector):
doi.org/10.5281/zenodo.20319421
- **P48 — The Complete GTE Framework** (main synthesis monograph):
doi.org/10.5281/zenodo.20560550

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

- [1] Nova Spivack. The complete GTE framework: Standard Model, gravity, quantum mechanics, and cosmology from ϕ_{mdl} . Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.20560550>. <https://doi.org/10.5281/zenodo.20560550>. Preprint.
- [2] Nova Spivack. Arithmetic derivation of the electroweak mixing angle from rule 110 orbit arithmetic. Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.20319413>. Paper P31 in the UGP Physics series. <https://doi.org/10.5281/zenodo.20319413>.